

Laboratory Preparation, Pin Mapping and Configuration

(Note: The base addresses and configurations below assume the STM32F4xx series, which is standard for Cortex-M4 educational boards.)

1. Register Map (STM32F4xx Base Addresses)

This table outlines the starting memory addresses for all required hardware peripherals.

Peripheral	Bus	Base Address	Purpose in Project
GPIOA	AHB1	0x4002 0000	SPI lines, Motor PWM LED, UART TX
GPIOB	AHB1	0x4002 0400	Hallway Call Buttons, Floor Sensors
GPIOC	AHB1	0x4002 0800	Internal Cabin Floor Buttons
GPIOD	AHB1	0x4002 0C00	Emergency Stop Button
RCC	AHB1	0x4002 3800	Clock enablement for all peripherals
EXTI	APB2	0x4001 3C00	External Interrupt controller for asynchronous inputs
NVIC	Core	0xE000 E100	Interrupt prioritization (Emergency Stop highest)
SPI1	APB2	0x4001 3000	Full-Duplex Inter-Processor Communication (IPC)
USART2	APB1	0x4000 4400	500ms Telemetry logging to PC
TIM2	APB1	0x4000 0000	PWM generation for motor simulation LED
TIM3	APB1	0x4000 0400	Hardware scheduler (Triggers SPI sync and UART TX)
DMA1	AHB1	0x4002 6000	(Bonus) Zero-CPU-overhead UART transmission

2. PWM Math (10kHz Frequency)

To achieve smooth LED dimming representing motor speed (0%, 20%, 100%), we configure Timer 2 (TIM2).

- Assumption:** The System Clock (SYSCLK)/Timer Clock is running at 16 MHz (Default HSI).

- **Target Frequency:** 10,000 Hz (10 kHz).
- **Formula:** $F_{\text{PWM}} = \frac{F_{\text{Timer}}}{(PSC + 1) \times (ARR + 1)}$

Calculations:

To make controlling the Duty Cycle easy, we set the Auto-Reload Register (ARR) to 99. This gives exactly 100 steps (0 to 99), so $ARR + 1 = 100$.

$$10,000 = \frac{16,000,000}{(PSC + 1) \times 100}$$

$$10,000 \times 100 = \frac{16,000,000}{(PSC + 1)}$$

$$1,000,000 = \frac{16,000,000}{(PSC + 1)}$$

$$PSC + 1 = 16 \implies \mathbf{PSC = 15}$$

Result:

- **Prescaler (PSC)** = 15
- **Auto-Reload (ARR)** = 99
- *To achieve 20% speed:* Set Capture/Compare Register (CCR) = 20 .
- *To achieve 100% speed:* Set Capture/Compare Register (CCR) = 100 .

3. Packet Definition (8-Byte SPI Frame)

This is the fixed data structure transmitted between the Master and Slave every 50ms.

Byte Index	Field Name	Description	Example Value
Byte 0	Header	Start of frame identifier mandated by rubric.	0xA5
Byte 1	Current Floor	The physical floor the elevator is currently at.	0x01 to 0x04
Byte 2	Target Floor	The destination floor.	0x01 to 0x04 (or 0x00 if IDLE)
Byte 3	Current State	FSM State (0=IDLE, 1=UP, 2=DOWN, 3=DOORS)	0x00 to 0x03
Byte 4	System Flags	Bit 0: Emergency Stop, Bit 1: Door Blocked, etc.	0x00 (Normal)
Byte 5	Reserved	Padding for future features or alignment.	0x00

Byte Index	Field Name	Description	Example Value
Byte 6	Reserved	Padding for future features or alignment.	0x00
Byte 7	Checksum	XOR calculation of Bytes 0 through 6 to detect glitches.	(Calculated)

Checksum Logic: $\text{Byte 7} = \text{Byte 0} \wedge \text{Byte 1} \wedge \text{Byte 2} \wedge \text{Byte 3} \wedge \text{Byte 4} \wedge \text{Byte 5} \wedge \text{Byte 6}$

4. Master Pin & Configuration Table (Proteus Simulation)

This table combines the logical requirement, the physical STM32 pin, how it connects in Proteus, and the required C-code configuration.

System Function	MCU Pin	Proteus Interface	Code Configuration (GPIO / Peripherals)
SPI Chip Select (CS)	PA4	Connect to Slave PA4 & Debugger SS	Master: Output Push-Pull / Slave: Input Floating
SPI Clock (SCK)	PA5	Connect to Slave PA5 & Debugger SCK	Alternate Function (AF05)
SPI Master In (MISO)	PA6	Connect to Slave PA6 & Debugger SDO	Alternate Function (AF05)
SPI Master Out (MOSI)	PA7	Connect to Slave PA7 & Debugger SDI	Alternate Function (AF05)
Motor Simulation LED	PA0	LED-GREEN via Resistor	Alternate Function (TIM2_CH1)(PWM Output)
UART Telemetry (TX)	PA2	Virtual Terminal RXD	Alternate Function (AF07)(USART2 TX)
Floor 1 Sensor	PB1 ₂	LOGICSTATE	Input (Pull-Down or Floating)
Floor 2 Sensor	PB1 ₃	LOGICSTATE	Input (Pull-Down or Floating)
Floor 3 Sensor	PB1 ₄	LOGICSTATE	Input (Pull-Down or Floating)
Floor 4 Sensor	PB1 ₅	LOGICSTATE	Input (Pull-Down or Floating)

System Function	MCU Pin	Proteus Interface	Code Configuration (GPIO / Peripherals)
Cabin Button 1	PC6	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge trigger
Cabin Button 2	PC7	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge trigger
Cabin Button 3	PC8	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge trigger
Cabin Button 4	PC9	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge trigger
Emergency Stop	PD11	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge, Highest NVIC Priority
Hallway Call: U1	PB0	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge (Master Only)
Hallway Call: D2	PB1	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge (Master Only)
Hallway Call: U2	PB2	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge (Master Only)
Hallway Call: D3	PB3	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge (Master Only)
Hallway Call: U3	PB4	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge (Master Only)
Hallway Call: D4	PB5	BUTTON to Ground	EXTI Input, Internal Pull-Up, Falling Edge (Master Only)

(Note: The Slave MCU setup is identical to this table, except you omit the PA2 UART connection and the PB0-PB5 Hallway Calls).